

Regeneration as Field-Structured Repair

Bioelectric Architecture, Wound Field Dynamics, and the Voltage Logic of Morphogenetic Restoration

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Abstract

Regeneration is commonly framed as either a genetic readout problem, which regeneration genes does this organism have?, or a stem cell deployment problem, does this organism retain the progenitor populations needed to rebuild lost tissue? Both framings capture something real, but both miss what the evidence increasingly demands we acknowledge as the primary control layer. Regeneration is, at its core, a field restoration problem. Multicellular bodies maintain a bioelectric field architecture, a spatially organized map of transmembrane voltages, ion current flows, gap junction network topologies, and electrical domain boundaries, that encodes positional identity, axial polarity, organ boundaries, and growth termination targets. Injury collapses this architecture locally and sometimes regionally. What follows is not primarily a molecular cascade that happens to produce tissue reconstruction; it is a sequence of field restoration operations. Wound field formation provides the first spatial signal. Polarity re-establishment resolves the symmetry-breaking question of what to build and in which orientation. Voltage domain encoding specifies fine-grained positional identity. Gap junction network reconstruction re-establishes the electrical integration required to maintain domain coherence. Electrotaxis-driven cell migration populates the repair site. And developmental template replay, driven by the restored field architecture, rebuilds the missing structure by recapitulating a compressed developmental program. This paper walks through each mechanism in mechanistic order, drawing on evidence from planarians, axolotls, *Xenopus* frogs, and human wound biology.

The central argument is that the degree of regenerative competence an organism possesses is largely, and perhaps primarily, a function of how accurately and how durably it can restore bioelectric field coherence after injury. This framework does not replace molecular biology; it situates molecular biology within an explanatory structure that actually accounts for the spatial and temporal ordering of regenerative events.

SECTION 1

THE BODY IS AN ELECTRICAL MACHINE (AND ALWAYS HAS BEEN)

Every living cell in a multicellular body maintains a voltage difference across its plasma membrane. This is not a side effect of being alive, it is one of the most fundamental features of cellular existence, predating the evolution of complex tissues by hundreds of millions of years. The **resting transmembrane voltage** (V_{mem}) is set by the asymmetric distribution of ions across the plasma membrane, driven by the active work of ion pumps and the passive flow of ions through channels and transporters. The **Na⁺/K⁺-ATPase** pumps three sodium ions out for every two potassium ions pumped in, generating both a chemical gradient and a charge imbalance. The **H⁺-ATPase** (proton pump, also called V-ATPase in its vacuolar form) extrudes protons, contributing additional electrogenic potential. Chloride channels, potassium leak channels, and a host of regulated ion transporters further sculpt the final V_{mem} that a given cell maintains at rest in a given tissue context. The result is a plasma membrane sitting at a measurable electrical potential, typically between -10 mV and -90 mV relative to the cell exterior, depending on cell type and functional state.

What makes this relevant to regeneration, and what took biology an embarrassingly long time to fully appreciate, is that V_{mem} is not merely a byproduct of metabolic housekeeping. It is a **regulatory variable**. The voltage across the plasma membrane directly controls whether a cell proliferates, differentiates, migrates, or remains quiescent.

This control is not indirect or metaphorical. Voltage-gated ion channels couple V_{mem} to intracellular calcium dynamics; membrane potential modulates the activity of enzymes that depend on intracellular pH; voltage changes drive conformational shifts in membrane receptors that alter their downstream signaling competence. V_{mem} is a signal in the most literal sense: information encoded in a physical quantity that the cell's machinery actively reads and responds to.

The voltage ranges associated with different cellular states are informative. Actively proliferating cells, including stem cells and cancer cells, cluster in the range of -10 to -30 mV, which is relatively depolarized compared to the cell exterior. Terminally differentiated, post-mitotic cells sit at -50 to -90 mV: hyperpolarized, quiet, committed to their identity. This is not a coincidence, and it is not merely correlational. When you artificially hyperpolarize cells that are in a proliferative state, you push them toward differentiation. When you artificially depolarize differentiated cells, you can drive re-entry into the cell cycle. The relationship between voltage and cell state is causal, bidirectional, and mechanistically grounded, not hand-waving.

The spatial dimension of all this emerges through **gap junctions**. Gap junctions are intercellular pores formed by the docking of two hemichannels, each composed of six connexin protein subunits, one contributed by each of the two adjacent cells. When the hemichannels dock, they form a continuous aqueous channel approximately 1.2 to 2 nanometers in diameter, connecting the cytoplasm of adjacent cells. Through these pores, ions flow down electrochemical gradients, small signaling molecules including inositol trisphosphate (IP_3), cyclic AMP (cAMP), and calcium ions (Ca^{2+}) transit between cells, and even some small charged metabolites and RNA species can pass. The critical consequence is that electrically, the tissue is not a collection of isolated cells, it is a network. Ion flows through gap junctions can be directional, driven by V_{mem} differences between coupled cells, generating intracellular ionic currents that carry positional information across tissue distances that no diffusible morphogen could span with equivalent speed or spatial resolution.

The result of all this coupling is that tissues maintain **standing voltage gradients**, spatial patterns of V_{mem} that vary systematically from one tissue region to another. These are not noise. They are organized, reproducible, and functionally meaningful. Individual cells experience not just their own V_{mem} but the electrical state of all the cells they are gap-junction coupled to, which means their behavior is influenced by the collective bioelectric state of their tissue neighborhood. A cell at the anterior tip of an embryo is electrically coupled into a different network context than a cell at the posterior end, and that difference in electrical context is part of how it knows where it is.

The most compelling early demonstration of this principle comes from *Xenopus laevis* embryos, where voltage-sensitive fluorescent dyes reveal reproducible V_{mem} maps across the body axis well before any tissue differentiation is histologically detectable. You can read the future anatomy, the prospective territories of the head, trunk, gut, and limb fields, in the present voltage pattern. Distinct voltage domains prefigure distinct anatomical territories with a consistency that is not compatible with random electrical noise. The pattern is conserved across embryos and predicts what will develop there. In other words, the voltage is doing the thinking before the genes even get involved in laying out the body plan.

This sets up the central argument of the paper with a logical inevitability. If the intact body is a bioelectric map, a spatially organized pattern of voltages, currents, and electrical domains encoding positional identity and growth states, then injury is not primarily a structural event or a molecular event. It is, first and most immediately, a map disruption event. A cut through tissue is a cut through the electrical architecture that organizes that tissue. And if that is true, then regeneration, the rebuilding of lost structures with correct form and function, is fundamentally a map reconstruction event. Everything else: the stem cell activation, the morphogen gradients, the transcription factor cascades, these are the molecular machinery executing a plan whose coordinates are written in bioelectric field structure. The question is whether the body can rewrite the map, and how.

SECTION 2

INJURY TRIGGERS FIELD COLLAPSE

When tissue is cut, the first thing that happens is not a growth factor release. It is not an immune cell infiltration. It is not a gene expression change. The first thing that happens is electrical, and it happens at the speed of electricity, which is to say, effectively instantaneously. A physical breach in an epithelial barrier is simultaneously a breach in the electrical architecture that barrier maintains, and the consequences propagate through the surrounding tissue faster than any molecular signal can move.

The key concept here is the **transepithelial potential** (TEP): the standing voltage difference maintained across an intact epithelial sheet, with the luminal or apical surface typically negative and the basal surface positive, ranging from 10 mV to 100 mV depending on the tissue. This TEP is not incidental to epithelial function, it is actively maintained by the coordinated activity of ion pumps and channels expressed in spatially segregated membrane domains. In the skin, the apical membrane of keratinocytes preferentially expresses sodium channels (particularly ENaC), while basolateral membranes house the Na⁺/K⁺-ATPase. The resulting asymmetric transport creates a steady net positive charge accumulation on the basal side relative to the luminal side. This is the body's way of establishing and maintaining a local electrical potential that will turn out to be critically important the moment something goes wrong.

When the epithelium is breached, the barrier that maintained this potential is suddenly gone at the wound site. Ions leak freely across the open circuit. The pumps that worked against a resistive barrier are now pumping against an open circuit, and the local TEP collapses to near zero at the wound edge. But the surrounding intact epithelium is still maintaining its TEP. The result is a lateral voltage difference between the still-intact tissue and the collapsed wound site, and voltage differences drive current flows. A **wound electric field** forms, pointing toward the wound center (from high potential to low potential), with field strengths in human and animal skin typically measured at 40 to 200

mV/mm in the first few hundred microns from the wound edge. To put that in context: that is an enormous gradient. The fields driving neuronal signaling in the brain are in a similar range. This is not a subtle perturbation.

This wound electric field is emphatically not an epiphenomenon. It is the first spatial signal that the surrounding tissue receives indicating that damage has occurred and precisely where it is. Every cell in the vicinity of the wound is simultaneously informed, by the direction and magnitude of this electrical gradient, that the wound is located in a specific direction at a specific distance. No diffusible signal can convey that information this fast. Growth factors like EGF and PDGF will follow, chemokines will be secreted, cytokines will be released, but the electric field arrives first, establishes the spatial coordinate, and begins organizing the cellular response before the molecular signals can propagate more than a cell diameter or two.

Simultaneously with TEP collapse, injury triggers **calcium waves** that propagate outward from the wound edge through a combination of gap junction-mediated calcium transit and paracrine ATP release activating purinergic receptors on neighboring cells. These calcium waves are not independent of the bioelectric changes, they are coupled to them. Intracellular calcium concentration directly modulates V_{mem} through calcium-activated potassium channels (BK channels and SK channels), creating a feedback loop: the wound-induced calcium wave changes V_{mem} in surrounding cells, which changes their ion channel behavior, which further propagates the electrical disturbance outward. The calcium wave and the bioelectric perturbation are co-propagating disturbances in the same coupled system.

The depolarization zone at the wound site is a measurable, imageable bioelectric signature. In planarian flatworms, voltage-sensitive dyes allow real-time imaging of the electrical state of the tissue after cutting, and you can watch the depolarization zone form within minutes. The wound site is electrically distinct from the surrounding tissue within a timeframe that precedes any visible wound healing response. The zone has defined boundaries, and those boundaries matter: they will later determine where the wound field transitions to intact

tissue field, and that transition zone is where a great deal of the interesting regenerative action will happen.

A mechanistically important contributor to wound field dynamics is the production of **reactive oxygen species** (ROS) at the wound site. Injury activates NADPH oxidase enzymes in damaged and immediately neighboring cells, generating superoxide and its downstream products including hydrogen peroxide. This ROS burst is not just inflammatory signaling, it directly activates specific ion channels, particularly **TRPM2** (Transient Receptor Potential Melastatin 2), a calcium-permeable cation channel that is gated by oxidative stress. TRPM2 opening drives depolarization and calcium influx in the cells surrounding the wound, amplifying the initial bioelectric signal and helping to sharpen the spatial boundary of the wound field. ROS are, in this light, part of the bioelectric signaling cascade, not just damage markers, but active participants in defining the electrical geography of the injury zone.

Axolotl limb amputation provides one of the most studied and informative examples of wound field formation and its subsequent dynamics. Within the first hour after amputation, a characteristic ion current pattern can be measured using vibrating probe electrodes placed in the solution just outside the wound surface: a strong outward current of positive ions (predominantly Na⁺ and K⁺) flows out from the wound face. This current, driven by the collapse of the limb's standing bioelectric architecture, persists and evolves over the following days as the wound epidermis closes and the blastema begins to form. The current pattern changes as the regenerative program advances, and, critically, perturbations to this current pattern produce predictable perturbations in regenerative outcome. The wound field is not background noise. It is the regenerative program's first written line.

SECTION 3

WOUND FIELDS FORM INSTANTLY, AND THEY CARRY INFORMATION

The wound electric field that forms in the minutes following injury is not uniform noise. It is structured. The field points toward the wound center, with field strength greatest right at the wound margin and falling off with distance from the wound edge in a gradient that reflects the geometry of the ionic leak and the resistance of the surrounding tissue. This spatial structure is immediately and automatically informative: it gives every cell in the wound vicinity a directional vector pointing toward the damage site, along with a distance estimate encoded in the local field magnitude. Before any cell has received a diffusible signal, before any growth factor has been secreted and bound to a receptor, the electric field has already established a coordinate system centered on the wound. This is the first piece of spatial information the regenerating tissue operates on, and it is delivered at electromagnetic propagation speeds.

Cells detect electric fields through several distinct and complementary mechanisms, and understanding these mechanisms is essential to understanding why the wound field is biologically effective rather than just physically present. The first and perhaps most important mechanism is the electrophoretic redistribution of charged membrane components. Membrane proteins that carry a net charge, including the **epidermal growth factor receptor** (EGFR) and various integrin subunits, are mobile within the plane of the lipid bilayer. In the presence of a lateral electric field, these charged proteins are driven by electrophoresis toward one face of the cell. In cathodal-migrating cell types (those that move toward the negative pole of the field, which is the wound center in most wound geometries), EGFR accumulates on the cathodal-facing membrane within minutes of field application. This receptor asymmetry creates a signaling asymmetry before any directed cell movement has occurred: the cathodal face of the cell has more EGFR, activates more downstream signaling, and is primed for protrusion.

The downstream consequence of EGFR redistribution is a spatial asymmetry in **PI₃K/PTEN** activity. PI₃K (phosphatidylinositol 3-kinase) is activated by EGFR and generates PIP₃ (phosphatidylinositol-3,4,5-trisphosphate) at the inner leaflet of the plasma membrane. PTEN is the lipid phosphatase that degrades PIP₃. In a cell exposed to a wound electric field, the cathodal redistribution of EGFR drives PI₃K activation preferentially on the cathodal face, generating a PIP₃ gradient that points the cell toward the wound. This PIP₃ gradient is the same gradient that drives chemotactic polarity in response to diffusible signals, the electric field is commandeering the same cellular polarity machinery through a different upstream input. The result is that cells in the wound field are already polarized, already biased toward the wound, before a single actin filament has been polymerized in the direction of migration.

Cells also respond to electric fields through **voltage-sensitive calcium channels (VSCCs)**. Field exposure causes asymmetric calcium influx through VSCCs on the cathodal-facing membrane, creating a cytoplasmic calcium gradient within the cell that closely mirrors the extracellular field orientation. This intracellular calcium gradient activates **Rac1** and **Cdc42** at the calcium-high (cathodal) face, these small GTPases drive actin branching and lamellipodia formation through the **Arp2/3** complex. Simultaneously, at the anodal face where calcium influx is lower, **RhoA** activity is elevated, driving actomyosin contractility and retraction. The result is the classic protrusion-at-the-front, retraction-at-the-rear cell polarity that underlies directional migration, established by an electric field-driven calcium asymmetry, not by a diffusible chemotactic gradient.

The wound field is self-amplifying up to a point, and the mechanism of self-amplification involves the migrating cells themselves. As keratinocytes at the wound edge begin to polarize and migrate in response to the field, they actively maintain and reinforce the local electrical architecture. They do this by preserving their tight junction complexes on lateral and basal membrane domains, maintaining the electrical barrier function of those surfaces, while their apical surfaces face the wound environment. This geometry means that as the wound epidermis advances, it maintains the ion transport asymmetry required to sustain

the TEP, thereby sustaining the wound field that is driving its own migration. It is a self-reinforcing cycle: the field drives migration, the migrating cells maintain the field, and this continues until wound closure resolves the electrical discontinuity that started the whole process.

Corneal wound healing provides one of the cleanest demonstrations of wound electric field function in a physiologically accessible system. The corneal epithelium maintains a TEP of about 25 to 40 mV, and corneal wounds generate a lateral wound electric field of approximately 40 mV/mm at the wound margin, entirely within the physiologically generated, endogenous range. When this field is experimentally neutralized by short-circuiting the corneal epithelium (reducing the TEP to near zero with pharmacological agents that dissipate the ion gradients), wound healing is severely impaired even though all the relevant growth factors, cytokines, and cell populations are still present and intact. The biochemistry is fine. The cells are there. The molecular signals are present. But without the electric field, the system cannot coordinate directional migration effectively, and healing slows dramatically. This is a clean dissociation: it demonstrates that the field is not just a correlate of healing but a necessary organizer of it.

There is a temporal argument worth making explicit here, because it matters for understanding causality. Wound electric fields form at the speed of ion flow through a resistive medium, essentially instantaneously on biological timescales, within milliseconds of injury. The fastest diffusible wound signal is ATP released from damaged cells, which can spread over hundreds of microns in seconds to minutes. Growth factor gradients, cytokine gradients, and chemokine gradients require minutes to hours to establish meaningful spatial structure. This temporal ordering is not an accident: the electric field is upstream of the molecular signals not just logically, but chronologically. It provides spatial coordinates before any molecular signal has organized enough to be read. The cells at the wound margin are already electrically oriented, already polarized, already primed to migrate, before the first growth factor molecule has diffused a cell diameter. This is why

bioelectric signals lead the wound response, and why studying only the molecular layer gives a picture of regeneration that is missing its first chapter.

The wound field also establishes the voltage gradient that will become the substrate for subsequent polarity encoding, the critical downstream step in which the regenerating system decides not just where the wound is, but what to build there. This is the transition from damage detection to morphogenetic instruction, and it runs entirely on the bioelectric infrastructure that the wound field has established. The wound field is not just a migration cue. It is the first draft of the new anatomical plan.

SECTION 4

POLARITY RESTORATION, DECIDING WHAT TO BUILD AND WHERE

After a wound field establishes itself and begins driving the initial cellular response, the regenerating system faces a question that is more profound than it might initially appear: what should grow back, and in what orientation? This is the polarity problem, and it is not trivial in the slightest. A sheet of keratinocytes migrating to close a skin wound does not need to solve it, the wound closes, and the structural information of the surrounding intact tissue provides sufficient context. But a planarian flatworm cut in half faces a genuinely hard problem: the anterior fragment needs to regenerate a new tail at its posterior-facing wound, and the posterior fragment needs to regenerate a new head at its anterior-facing wound. Both fragments contain cells with identical genomes. Both wound faces see the same chemical environment, the same seawater, the same released intracellular contents from damaged cells. Whatever resolves the asymmetry between "grow a head here" and "grow a tail here" cannot be in the molecular injury signal, because that signal is the same. It has to be in the state of the tissue. And it is, the state of the bioelectric field.

Within minutes of amputation in planarians, a measurable voltage difference develops between the two wound faces. The anterior-facing wound blastema (the one that will form a head) is measurably more hyperpolarized than the posterior-facing wound blastema (the

one that will form a tail). This is not a small or ambiguous difference, it is a clear, reproducible, measurable electrochemical distinction between the two wound surfaces, arising from differences in the ion channel and pump composition of the cells at each end. The anterior tissue and the posterior tissue have different bioelectric identities in the intact animal, and those identities are partially maintained at the wound face even after amputation. The polarity decision is not made from scratch, it is read from a voltage state that pre-exists the cut.

The mechanism linking this early voltage asymmetry to gene expression and tissue identity is increasingly well characterized. At the hyperpolarized anterior wound face, the voltage state favors the expression and activity of **Notum**, a secreted Wnt antagonist. Notum expression inhibits Wnt pathway signaling in cells at the anterior blastema, suppressing **β-catenin** nuclear translocation and pushing those cells toward head identity, in planarians, head identity is the Wnt-low state. At the depolarized posterior wound face, the voltage state promotes Wnt pathway activity. β-catenin is active, target genes for tail identity are expressed, and the posterior blastema commits to producing a tail. The bioelectric state comes first. The gene expression asymmetry follows from it, not the other way around.

The proof-of-concept experiments in this area are frankly stunning, and they deserve to be stated plainly. If you pharmacologically depolarize both wound faces immediately after amputation in planarians, using ionophores like **nigericin** or **monensin** that dissipate the potassium gradient and force cellular depolarization, both wound faces now see the same depolarized voltage state that normally characterizes only the posterior blastema. Both blastemata then express tail-identity genes. The result is a worm with two tails and no head. Conversely, if you push both wound faces toward hyperpolarization using ouabain to block Na⁺/K⁺-ATPase in a way that paradoxically increases membrane potential in certain cell types, or by expressing specific potassium channels, both blastemata can be driven toward head identity and you get a two-headed worm. You have completely overwritten the animal's body plan axis using a voltage manipulation applied in the minutes immediately following amputation, before any cell has divided, before any significant morphogenetic

movement has occurred. That is not subtle. That is the bioelectric field dictating to the genome what body plan to execute.

This kind of experiment demonstrates something that is conceptually important and not yet sufficiently absorbed into mainstream regeneration biology: polarity is not hardwired in the sense of being irreversibly fixed by genomic sequence. It is an emergent property of the bioelectric state of the tissue, and it can be rewritten if you know the code. The genome contains the hardware to build both a head and a tail, every cell has that hardware. The bioelectric state determines which program runs in which location. Calling polarity "genetic" in the sense that it is directly and irrevocably specified by the DNA sequence is true but superficial, it is like saying the output of a computer is "hardware-determined" because the transistors are made of silicon. Technically correct, entirely misleading about where the interesting decisions are made.

In vertebrate limb regeneration, polarity operates along multiple axes simultaneously, proximal-distal (shoulder to fingertip), anterior-posterior (thumb side to pinky side), and dorsal-ventral (back-of-hand to palm). All three axes need to be correctly specified in the regenerating blastema, and all three show bioelectric correlates that precede and appear to direct the molecular patterning. In axolotl limb regeneration, the blastema at the amputation site has a characteristic voltage profile that is distinct from the surrounding stump tissue, and that profile evolves over the course of regeneration in a way that tracks the progressive specification of the missing limb segments. When this profile is disrupted with specific ion channel blockers, particularly blockers of the potassium channels that contribute to blastema hyperpolarization, regenerative defects follow: missing distal segments, duplicated elements, and in some cases, partial polarity reversals. The blastema is reading a voltage pattern to know where it is in the proximal-distal axis and what to build next, and you can scramble that readout by scrambling the voltage pattern.

The key conceptual insight this section is building toward is the following: polarity in regeneration is not determined by which transcription factors happen to be active at the moment of injury, as a snapshot of the cell's prior commitment. It is determined by which

bioelectric attractor state the wound region settles into during the minutes to hours following injury, and that attractor state then drives the gene expression program that executes the corresponding body plan. The bioelectric state is the master variable. Gene expression is downstream. This is not to say that the genes are unimportant, they are absolutely essential, but the spatial instruction that tells the genes what to do in which location is written in voltage, not in the genome itself. The genome is the recipe book; the bioelectric field is the chef deciding what to cook and where to serve it.

SECTION 5

VOLTAGE DOMAINS, THE BODY'S ZIP CODE SYSTEM

Polarity is a coarse-grained signal. Head versus tail, proximal versus distal, anterior versus posterior, these are the large-scale coordinates, and they are absolutely necessary, but they are nowhere near sufficient to specify the thousands of distinct cell identities and tissue structures that a regenerating limb or organ must produce. A regenerating salamander limb needs to know not just that it is building a distal limb, but that it is building a third digit, in the posterior-dorsal quadrant, at the level of the second phalanx. That kind of fine-grained positional identity requires a higher-resolution encoding system. Voltage domains are that system. They are the body's zip code: your V_{mem} , in the context of the V_{mem} of your immediate electrical neighbors, encodes your address within the tissue with a specificity that coarse-grained polarity gradients alone cannot provide.

A **voltage domain** is a coherent region of tissue in which cells share a similar V_{mem} profile and are electrically coupled to one another through gap junctions. The boundary between adjacent voltage domains is marked by a transition in V_{mem} , often relatively sharp, maintained by differences in ion channel and pump expression between the two domain cell populations. These transitions are not passive, they are actively maintained by the differential ion transport activity of the cells on either side. The result is a tissue landscape punctuated by electrical boundaries, not unlike the way a country is divided into postal zones with distinct codes separated by defined borders.

Voltage domain mapping in *Xenopus laevis* embryos using voltage-sensitive fluorescent dyes has produced some of the most compelling evidence for this model. Distinct Vmem domains appear in early *Xenopus* embryos and reliably prefigure future organ territories before organ-specific transcription factors are expressed. The territory that will become the eye field has a characteristic Vmem distinct from its neighbors. The territory that will become the brain, the gut, the heart, each has its own electrical signature, and these signatures are present before the tissue has differentiated histologically, before marker genes have turned on in those territories. The electrical map precedes the molecular map. The zip code system is written before the addresses are built.

What makes this causal rather than merely correlational is the ability to perturb voltage domains and observe corresponding changes in organ identity. When researchers misexpress specific ion channels in defined regions of the *Xenopus* embryo, using injected mRNA that is expressed only in targeted cell populations, they can shift the Vmem of a targeted region toward the characteristic voltage of a different domain. Depolarizing the prospective eye field disrupts eye development in a predictable, dose-dependent way. More dramatically, hyperpolarizing a non-eye territory to the Vmem level characteristic of the eye field can sometimes induce the formation of ectopic eye-like structures, not just ectopic expression of eye marker genes, but partial eye structures forming outside the normal eye territory, at locations specified by an experimentally imposed voltage domain. This is extraordinary: you imposed a voltage pattern and the tissue tried to build the organ normally specified by that voltage pattern, even in an anatomically incorrect location. The gene networks followed the voltage cue.

In regeneration specifically, voltage domain re-establishment is a critical and temporally early step that precedes and enables correct tissue patterning. The **blastema**, the mass of dedifferentiated, proliferating cells that forms at an amputation site and gives rise to the regenerating structure, has a characteristic and evolving voltage profile. Early in blastema formation, the Vmem profile is relatively undifferentiated; as regeneration proceeds, distinct voltage subdomains emerge within the blastema that prefigure the patterning of

distinct tissue types and positional identities in the regenerating structure. The emergence of these electrical subdomains precedes the expression of the tissue-specific transcription factors and signaling molecules that ultimately specify those tissues. The field is ahead of the biology, leading the anatomy rather than reflecting it.

Gap junctions are essential to voltage domain maintenance, and this cannot be emphasized strongly enough. Voltage domains are not just the summed activity of independently regulated cells, they require the cells to be electrically integrated through gap junction coupling so that the domain can maintain a coherent, uniform voltage across its extent. When gap junction function is disrupted pharmacologically, using connexin-blocking agents like carbenoxolone or specific peptide inhibitors, voltage domain coherence is lost even when all the individual cells are still expressing the correct ion pumps and channels. The domain boundaries dissolve, the V_{mem} pattern becomes noisy and heterogeneous, and patterning defects follow. The critical point is that these defects occur even when the molecular machinery of patterning is intact. Transcription factors are present. Morphogen sources are present. But without the electrical domain structure that integrates cells into coherent territories, the developmental program cannot execute correctly. Electrical integration is not optional, it is part of the specification mechanism.

The **V-ATPase proton pump** deserves specific attention here, because it is one of the primary writers of voltage domain identity in regenerating systems. V-ATPase drives proton extrusion across the plasma membrane in an electrogenic fashion: each proton pumped out carries a positive charge out of the cell, contributing to membrane hyperpolarization. Differential expression and activity of V-ATPase across tissue regions is a major driver of the V_{mem} differences between voltage domains. In planarians, V-ATPase activity is high in the posterior-facing blastema and contributes to the depolarization that specifies tail identity at that wound face. When V-ATPase is pharmacologically inhibited specifically at the posterior blastema, using targeted delivery of bafilomycin A_1 , that blastema shows a shift in V_{mem} toward hyperpolarization, and correspondingly shifts its gene expression toward head identity. A proton pump. Dictating head versus tail fate.

Through voltage. This is the kind of mechanism that makes you stop and appreciate just how different the body's information processing logic is from anything we initially assumed.

The critical conceptual frame to hold is this: by the time any transcription factor gradient or morphogen gradient is detectable by in situ hybridization or immunofluorescence in a regenerating tissue, the voltage domain pattern has been in place for hours. The electrical map is the first draft. The molecular map is the second draft, written in response to the first. When regenerative medicine asks why introducing the "right" transcription factors or morphogens into a wound sometimes fails to produce regeneration, this is part of the answer: the molecular layer is trying to write without the electrical template to write on. The missing piece is often not a molecule. It is the field architecture into which the molecule is supposed to be read.

SECTION 6

GAP JUNCTIONS, REBUILDING THE NETWORK

Voltage domains, as established in the previous section, are not maintained by individual cells acting in isolation. They require physical electrical coupling between cells, and that coupling is provided by gap junctions. It is worth being specific about the molecular biology here, because the details matter for understanding how regeneration can go right and how it can go catastrophically wrong. Gap junctions in vertebrates are formed by proteins of the connexin family, with 21 different **connexin** (Cx) isoforms encoded in the human genome. Each connexin isoform has distinct conductance properties, distinct selectivity for different ion species and small molecules, and distinct regulation by phosphorylation, pH, and calcium. The gap junction channel that two cells share is not generic, its properties depend on which connexin isoforms are expressed on each side, and different isoform combinations produce channels with meaningfully different physiological characteristics. Which connexins are expressed in a tissue at a given developmental or regenerative stage

determines what signals can flow through the gap junction network, and therefore determines what information the network can transmit.

In intact tissue, the gap junction network creates something approaching an electrical syncytium: a tissue-wide network in which ion flows can be directional and information-bearing, driven by V_{mem} gradients between connected cell populations. Through this network, the positional information encoded in voltage domains is continuously shared and equilibrated among coupled cells, maintaining the coherence of those domains against local perturbations. A single cell that receives a depolarizing signal from a local stimulus, a mechanical stress, a paracrine signal, is partially buffered against that perturbation by its electrical coupling to its neighbors, which maintain the domain V_{mem} and pull the perturbed cell back toward the domain state. Gap junction networks act as electrical buffers that stabilize voltage domain identity against noise. This is not a trivial function, it is what makes the bioelectric code robust and readable rather than a noisy fluctuation.

Injury disrupts the gap junction network at and near the wound, and this disruption is not merely collateral damage to be repaired along with everything else. The disruption actively and immediately alters the bioelectric state of the surrounding tissue. Cells that were integrated into a coherent voltage domain suddenly lose some of their electrical neighbors, the coupling network develops holes, and the voltage domain boundary near the wound becomes less well-defined. This loss of electrical coherence near the wound is part of what establishes the transition zone between the wound field and the intact tissue field, and it is part of why the wound field has the spatial structure it does. But it also means that one of the first orders of regenerative business is to rebuild the gap junction network, because without it, the voltage domains required for patterning cannot be stably maintained.

The temporal regulation of gap junction activity during wound healing is not a passive matter, it is tightly controlled, and the control is both adaptive and surprisingly nuanced. **Connexin 43** (Cx43), the most widely expressed connexin in mammals and one of the primary gap junction proteins in skin, is rapidly downregulated at the wound edge immediately following injury. This downregulation is adaptive: an open gap junction

network at a wound edge would allow runaway ion leak and calcium wave propagation to spread throughout the tissue, potentially destabilizing a far larger region than the wound itself. Cx43 downregulation acts as a circuit breaker, limiting the electrical disturbance to the wound vicinity and protecting the integrity of the voltage domain network in the surrounding intact tissue. This is elegant damage control, the system severs its own electrical connections near the wound to prevent the electrical consequences of injury from propagating network-wide.

What follows is equally important and equally controlled. As the wound epidermis closes, the acute phase of field collapse transitions to the constructive phase of field restoration. In regeneration-competent organisms, Cx43 expression comes back up at the healing wound as the wound closes, and gap junction coupling is progressively restored in the forming blastema tissue. This restoration is not just a return to baseline, the gap junction network that forms in the regenerating tissue has a specific topology that shapes how voltage domains develop in the blastema. Which cells are connected to which determines the electrical neighborhood of each cell, and electrical neighborhood is a primary determinant of Vmem state, which is a primary determinant of cell identity. Rebuilding the correct network topology is as important as rebuilding the correct cell composition.

Chronic non-healing wounds illustrate what goes wrong when this regulation fails. In diabetic foot ulcers and venous leg ulcers, the major classes of chronic wound, Cx43 regulation is persistently abnormal. Rather than being transiently downregulated at the wound edge and then restored as healing progresses, Cx43 is chronically elevated at the wound margin, maintaining high-conductance gap junction coupling in a zone where that coupling actively impedes the transition from the inflammatory to the proliferative phase of healing. The wound remains stuck in a state of persistent inflammation and ion dysregulation, unable to establish the bioelectric conditions required for productive regenerative signaling. This is a real clinical problem affecting millions of patients, and the current standard of care has no mechanistically targeted intervention at the bioelectric level. There are groups working on Cx43-inhibitory dressings and peptide-based

hemichannel blockers, and early results are encouraging, but the clinical adoption of bioelectric thinking in wound care remains frustratingly slow relative to the evidence base.

Different connexin isoforms have meaningfully different permeabilities to different signaling molecules, and this isoform specificity has regenerative implications. Cx43 is highly permeable to IP₃ and to ATP, making it effective at propagating calcium-mobilizing signals through the network. Cx26, by contrast, has higher permeability to cAMP. Cx30 has distinct conductance properties and is more sensitive to intracellular acidification. When the gap junction network in a regenerating tissue is reconstructed with a different isoform composition than the original, as can happen when regeneration is imperfect or when scar tissue forms, the informational content of the network changes. The channels are there, the cells are coupled, but the signals flowing through the network are different because the permeability profile is different. This is one reason why scar tissue, even when it has restored electrical connectivity, does not recapitulate the bioelectric function of the original tissue: the network is connected differently, carrying different signals, and therefore supporting different (and generally less organized) patterns of cellular behavior.

The practical conclusion is important and somewhat humbling for the field of regenerative medicine. You cannot think about regeneration as simply a problem of getting the right cells to the right place. You need those cells to be electrically integrated into a network with the correct topology, expressing the correct connexin isoforms, at the correct time in the regenerative sequence. Without that electrical integration, the voltage domains that carry positional information cannot be maintained, the developmental template cannot be stably read, and regeneration is fundamentally incomplete regardless of whether all the molecular players are present. The electrical integration is not the implementation layer. It is part of the specification layer. And we do not yet have good tools to engineer it deliberately *in vivo*.

SECTION 7

ELECTROTAXIS, ELECTRIC FIELDS DRIVING CELL MIGRATION

Once the wound field is established and the local bioelectric architecture has begun its first organizational operations, the field takes on an additional role that is more overtly mechanical: it drives cells toward the wound. This directional migration of cells in response to electric fields is called **electrotaxis**, or, equivalently, galvanotaxis when the term is applied to single-celled organisms. Electrotaxis is not a fringe phenomenon or an artifact of experimental overreach. It has been documented rigorously in keratinocytes, dermal fibroblasts, macrophages, neutrophils, epithelial cells from the cornea, gut, and respiratory tract, neural crest cells during development, and regenerating spinal cord axons, among others. The threshold field strength for detectable electrotactic response in these cell types is well within the range of endogenous wound fields: cells respond to fields as low as 10 to 25 mV/mm, and wound fields run from 40 to 200 mV/mm at the wound margin. The wound field is not just above the threshold, it is well into the responsive range for essentially every cell type that needs to contribute to wound healing.

The molecular mechanism of electrotaxis involves the same receptor redistribution and polarity establishment described in Section 3, but now coupled to the cytoskeletal effectors that actually drive cell movement. EGFR redistribution to the cathodal face of the cell, driven by electrophoresis, activates PI3K preferentially at that face, generating a PIP₃ gradient. PIP₃ recruits and activates **Rac1** and **Cdc42**, which in turn activate the **Arp2/3** complex, the actin nucleator responsible for branched actin network formation and lamellipodia extension. The cathodal face of the cell extends lamellipodia. The anodal face, where EGFR density is lower and PIP₃ is less abundant, shows elevated **RhoA** activity, driving actomyosin contraction and membrane retraction. This front-back polarity is established and maintained as long as the electric field persists, and the cell moves in the direction of the field vector. The machinery is the same machinery that drives chemotaxis,

electrotaxis is not a separately evolved response system, it is chemotaxis machinery being triggered by an electrical rather than a chemical upstream input.

The cytoskeletal response to wound-relevant electric fields is rapid. Cells can detect and respond to fields in the physiological range with measurable changes in lamellipodia dynamics within 5 to 10 minutes of field application. Directional bias becomes statistically clear within 15 to 30 minutes. This speed is important: wound electric fields form within seconds to minutes of injury, and the cellular response begins essentially as soon as the field is established. There is no delay while the system waits for diffusible signals to organize, the electric field provides an immediate, organized spatial stimulus that cells begin responding to in real time.

Electrotaxis also modulates cell migration speed independently of directionality. In many cell types, exposure to electric fields in the wound-relevant range (40 to 200 mV/mm) increases migration speed by 20 to 100% compared to field-free controls. This acceleration is not simply a result of the directional bias, cells in randomized field conditions (oscillating fields that reverse direction frequently) can show speed increases without directional bias, indicating that the field is acting as a general migration enhancer as well as a directional cue. The wound field is simultaneously a compass and an accelerant: it tells cells which way to go and makes them go faster. This dual action is probably important for the rapidity of wound closure, because the window in which the wound field is available as a guidance cue is finite, as the wound closes, the TEP is restored and the lateral field diminishes. The faster cells move during that window, the more ground they can cover before the signal fades.

Individual electrotaxis is interesting, but the more biologically significant phenomenon is **collective electrotaxis**. Cells at a wound edge do not migrate as individuals, they migrate as coherent sheets, maintaining cell-cell adhesions through E-cadherin and mechanically coupling their cytoskeletons through adherens junctions and the cortical actin network. This collective migration has a remarkable consequence for bioelectric guidance: the directional bias established in leading-edge cells by their exposure to the wound field is transmitted mechanically to follower cells through the adhesive network. Follower cells,

which may be tens or hundreds of microns from the wound edge and experiencing significantly weaker electric fields, receive a directional bias from their mechanically coupled leading-edge neighbors. A local electrical stimulus at the wound margin can propagate a directional migration response across the entire epithelial sheet, a near-global collective behavior driven by a local bioelectric signal. This is why wound closure is not limited by the density of cells immediately adjacent to the wound margin. The electrical signal, amplified by mechanical coupling, recruits the entire epithelial sheet into the closing movement.

The interaction between electrotaxis and chemotaxis deserves attention, because both are operating simultaneously in wound healing and they interact rather than simply adding. Cells in a wound field receive both an electrical directional signal and diffusible chemical signals including EGF, PDGF, SDF-1, and various other chemoattractants generated at the wound. In many cases, these signals are synergistic: both point toward the wound, and the cell integrates them through shared downstream effectors (PI₃K, Rac1, Cdc42). But the electric field establishes its signal faster than diffusible gradients can organize, meaning that the early phase of wound response is primarily electrically guided, while later phases may involve a more balanced integration of electrical and chemical signals. In conditions where the electric field and a chemotactic signal are in conflict, a situation that can be engineered experimentally by applying a field in the direction away from a chemokine source, cells often follow the electric field, at least at physiologically relevant field strengths. This suggests that in the hierarchy of guidance signals, the bioelectric cue holds a privileged position.

Long-range electrotaxis guides neural regeneration as well, and this dimension has direct clinical implications that are beginning to be translated. Regenerating axons in the spinal cord orient along endogenous electric fields that arise from the tissue architecture of the cord. Applied DC electric fields in the range of 25 to 100 mV/mm can redirect axon growth in severed spinal cord preparations, with the cathode attracting the growth cones of ascending sensory axons and, depending on fiber type, descending motor axons showing

different field preferences. Several clinical trials have applied DC electric fields to sites of spinal cord injury in humans, with some showing measurable improvements in functional outcomes. The field of electroceuticals, therapeutic interventions based on applied electric fields rather than drug molecules, is in its early clinical stages, but the mechanistic foundation is solid, and the results in animal models are increasingly compelling. Electrotaxis is not just a wound healing phenomenon. It is a general-purpose biological guidance system that the body uses across development and repair, and learning to deliberately deploy it is one of the more promising frontiers in regenerative medicine.

SECTION 8

FIELD-STRUCTURED GROWTH, REPLAYING THE DEVELOPMENTAL TEMPLATE

Everything discussed so far, wound field formation, polarity establishment, voltage domain encoding, gap junction network reconstruction, electrotaxis-driven cell migration, is ultimately prologue to the most demanding act in regeneration: actually rebuilding a complex, correctly patterned structure from scratch. In a truly regenerative organism, the restored bioelectric field architecture does not just heal the wound surface and restore a barrier. It initiates a process of **morphogenetic replay**: a compressed re-execution of the developmental program that originally built the missing structure, now running in the context of an adult organism, guided by the restored bioelectric template. Understanding why some organisms can do this and others cannot requires understanding what "replaying the developmental template" means mechanistically, and how the bioelectric field drives it.

In axolotls, the biochemical details of template replay have been studied extensively, and the picture that emerges is striking. A regenerating axolotl limb blastema re-expresses the major developmental patterning gene networks, **Shh** (Sonic Hedgehog) gradients re-establishing anterior-posterior digit identity, **Hox** gene expression patterns re-establishing proximal-distal segment identity, and **FGF** (fibroblast growth factor) signaling axes re-

establishing the apical ectodermal ridge equivalent that drives distal outgrowth. These are the same gene networks that operate during initial limb development in the embryo. But they are not activated randomly or in the wrong spatial arrangement, they are activated in positions that correctly correspond to the missing limb segments. The Shh source forms at the posterior edge of the blastema. The FGF-producing wound epidermis is at the distal tip. Hox gene expression domains form in the correct nested pattern along the proximal-distal axis. The spatial arrangement of the gene expression is correct. Something is telling these gene networks where to turn on, and that something is the bioelectric field architecture of the blastema, the voltage domain pattern that was established in the hours following amputation, before any of this gene expression was detectable.

The relationship between blastema bioelectric state and positional identity has been tested by transplantation experiments. If a mid-regeneration axolotl blastema is removed from its normal position and grafted to a different axial location, for instance, to the flank, where the bioelectric context differs from the normal limb position, the blastema often integrates into the new position's bioelectric field and shifts its patterning accordingly. It responds to the field it finds itself in, not purely to an internal positional memory it carries from its site of origin. The blastema is a pattern-following system, and the pattern it follows is bioelectric. This does not mean the blastema has no internal memory, late-stage blastemata are less plastic than early-stage ones, and there is a progressive commitment to positional identity as regeneration proceeds. But the early plasticity, driven by responsiveness to the local bioelectric field, is what allows regeneration to be robust to variations in the wound environment. The field coordinates the program; the cells execute it.

The mechanistic link between Vmem in blastema cells and transcription factor activity is not black-box or hand-wavy, it runs through specific, well-characterized signal transduction pathways. Voltage-dependent calcium channels (VSCCs) open and close in response to Vmem, and the calcium that flows through them activates calcineurin, a phosphatase that dephosphorylates and releases NFAT (Nuclear Factor of Activated T-cells) transcription factors for nuclear translocation. Different Vmem states produce

different patterns of VSCC activity, different calcium dynamics, and therefore different NFAT-regulated gene expression profiles. Separately, Vmem modulates intracellular cAMP levels through its effects on adenylyl cyclase activity and through the voltage sensitivity of certain HCN channels that regulate cAMP-gated ion flows. cAMP modulates **CREB** transcription factor activity, a major regulator of growth and differentiation genes. And the electrogenic activity of proton pumps creates an intracellular pH gradient that is regulated by Vmem, intracellular pH affects the activity of enzymes involved in chromatin remodeling, with direct consequences for gene accessibility and expression. These are not exotic pathways invented to explain bioelectric effects. They are core, conserved signal transduction cascades that happen to be sensitive to membrane voltage in ways that have been documented for decades but not assembled into a coherent picture until relatively recently.

Field-structured growth also enforces the size control that makes regeneration end at the right point. A regenerating limb must stop growing when it has reached the correct size and achieved the correct pattern, not before, not after, not a distorted version of the target. In developing and regenerating organisms, target morphology appears to be encoded as a stable bioelectric attractor state: a pattern of voltage domains, ionic current flows, and gap junction network topology that the system is continuously being driven toward and that, when achieved, reduces the signals that promote further growth. As a regenerating structure approaches its target size and pattern, the voltage domain map of the regenerating tissue progressively converges toward the voltage domain map of an intact tissue of that type. When convergence is achieved, when the electrical map matches the target map, growth-promoting signals (Wnt, FGF, BMP in growth-promoting contexts) are diminished by changes in their regulatory environment, and growth-inhibitory signals become dominant. The system has reached its target state and it knows it, because the bioelectric signal that would indicate deviation from the target state is no longer present.

Cancer, viewed through this framework, is instructively comprehensible as a failure of exactly this regulatory mechanism. Tumor cells are characteristically and persistently

depolarized, sitting at V_{mem} values of -15 to -30 mV, the same range as highly proliferative stem and progenitor cells. The gap junction coupling that would normally integrate them into growth-regulatory voltage domains is disrupted: Cx43 and other connexins are frequently downregulated in tumor cells, and the cells that are gap-junction coupled to them are often coupled through aberrant connexin isoforms or in abnormal network topologies. The tumor cells are not receiving the electrical stop signal that would normally arrive when they are integrated into a coherent, appropriately- V_{mem} domain. From a bioelectric standpoint, a tumor is a region of tissue that has lost field coherence and is consequently executing a permanent growth program, not because its genome is irreparably broken in every cell, but because its electrical integration with the tissue-level growth control network is broken. This framework makes several otherwise puzzling observations about cancer more legible: why tumors are often electrically isolated from surrounding normal tissue, why restoring gap junction coupling can suppress tumor growth in some contexts, and why the tumor microenvironment, which includes the bioelectric environment, is as important as the tumor cell's own genome in determining its behavior.

In organisms with limited regenerative capacity, adult frogs, mice, and humans, the developmental template replay is incomplete. The wound closes, scar tissue forms, and growth terminates. But the key point is that the voltage domain pattern restored by wound healing in these organisms is a *scar* pattern, a stable, low-information bioelectric state characteristic of fibrous connective tissue, not a pattern that encodes the target identity of the missing structure. The electrical template for the missing structure is never re-established in the wound site, and therefore the gene networks that would execute developmental template replay receive no coherent bioelectric instruction about what to build. The molecular machinery for regeneration is often present, adult mammalian cells retain the capacity to express Hox genes, Shh, FGF, and the other developmental patterning molecules when appropriate signals are provided. What is missing is the spatial organization of those signals, which depends on the voltage domain architecture that the wound healing response has failed to restore.

Proof that this constraint is not absolute comes from experiments in frogs. Tadpole tails normally regenerate after amputation, the bioelectric field architecture is well-maintained in this context. But this regenerative competence is progressively lost as the frog develops, and adult *Xenopus* frogs do not regenerate amputated limbs. However, when the bioelectric environment at the amputation site of a non-regenerating frog is pharmacologically manipulated, using channel-modulating drugs or applied electric fields to push the wound site toward a V_{mem} profile more characteristic of the regeneration-competent tadpole stage, partial regenerative responses can be induced. Not perfect limb regeneration, but measurable structural rebuilding that does not normally occur. The developmental template replay can be partially restarted by bioelectric intervention in an organism that has "lost" the ability to regenerate. The capacity is still in the genome. The missing piece was the electrical instruction. Provide the electrical instruction and the genome responds. This is both the most exciting result and the most important implication of the entire bioelectric framework.

SECTION 9

FIELD COHERENCE DETERMINES REGENERATIVE ABILITY

Having traced the full mechanistic sequence from injury through wound field formation, polarity encoding, voltage domain establishment, gap junction network reconstruction, electrotaxis-driven cell migration, and developmental template replay, we are now in a position to state the central thesis of this paper with the precision it deserves. The degree of regenerative competence that an organism possesses, its ability to rebuild lost structures with correct form, function, and spatial organization, is fundamentally a function of how accurately, how completely, and how durably it can restore **bioelectric field coherence** after injury. This is not a metaphorical claim. It is a mechanistic prediction that is increasingly supported by direct experimental evidence, and it has significant implications for how we think about the evolution of regeneration and the clinical possibility of enhancing it.

Field coherence, defined operationally, means that the bioelectric field architecture of the regenerating or healed tissue matches the target state: the voltage domain map, polarity gradients, gap junction network topology, and ion current patterns characteristic of an intact, fully patterned organism of that species. A regenerating tissue is field-coherent when its electrical map matches the electrical map of what it is supposed to be building. A tissue that has healed by scarring is field-incoherent in this sense: its electrical architecture matches the scar state, not the target tissue state. The organism has repaired the wound electrically, in the sense that the acute field collapse has been resolved and a stable bioelectric configuration has been achieved, but it has reached the wrong stable configuration, a scar attractor rather than a regeneration attractor.

Organisms that regenerate well are consistently characterized by rapid, accurate re-establishment of bioelectric patterning after injury. Planarians can regenerate complete, correctly patterned worms from small tissue fragments because their bioelectric patterning system is extraordinarily robust: the voltage domain map is re-established quickly and accurately from whatever fragment remains, the polarity asymmetry is re-established within minutes, and the gap junction network that integrates the regenerating tissue into a coherent electrical whole is rapidly restored. Axolotls re-establish blastema bioelectric architecture within hours of amputation, and this re-establishment tracks the progressive specification of the regenerating limb with remarkable fidelity. Hydra, which can regenerate from small clusters of dissociated cells, maintain a bioelectric identity in essentially any fragment that contains a sufficient number of gap-junction coupled cells, because the bioelectric information is distributed across the network rather than localized in any specific tissue region.

Organisms that regenerate poorly show a characteristic failure mode: wound closure and acute field recovery are relatively normal, but the fine-grained voltage domain pattern of the missing tissue is never re-established. Adult frogs close their wounds efficiently and restore TEP rapidly, but the bioelectric architecture of the wound site converges to a stable scar configuration rather than to the configuration of the missing limb segments. Adult

mice and humans show the same pattern, and the molecular correlates of this failure are identifiable. Mammalian scar tissue shows progressive downregulation of Cx43 and substitution with less conductive connexin isoforms in the scar core, reducing electrical integration and gap junction-mediated information flow. The ion channel expression profile in wound-site fibroblasts and myofibroblasts does not match the profile required to generate and maintain the voltage domain pattern of the target tissue. V-ATPase expression in the wound site does not reach the levels seen in regenerative blastema tissue. And critically, the stem and progenitor cells that infiltrate the wound site in mammals do not receive coherent bioelectric positional signals instructing them to adopt target tissue identities, because the voltage domain map of the target tissue is not present for them to read.

This framework illuminates several observations in regeneration biology that have been noted empirically for decades but lacked a coherent mechanistic explanation. Why is innervation required for limb regeneration in salamanders? Because peripheral nerves contribute to maintaining the bioelectric environment of the regenerating tissue, they carry ion currents, release neurotrophic factors that modulate ion channel expression in surrounding cells, and their axon membranes participate in the local gap junction network through coupling with Schwann cells. Denervated salamander limbs fail to form a proper blastema not because nerve fibers are needed as a structural scaffold, but because the bioelectric environment collapses without the neurally maintained ion current patterns. Why is regenerative competence in frogs lost at a specific developmental stage rather than gradually? Because that developmental stage corresponds to a transition in the bioelectric regulatory program, specific ion channels and gap junction isoforms are exchanged for adult-type variants, and the adult profile is less capable of restoring the regeneration-attractor bioelectric state after injury. Why do fetal mammalian wounds heal without scarring while adult wounds scar? Fetal wounds heal in an environment of high Cx43 expression and abundant amniotic fluid that maintains ionic homeostasis near the wound, the fetal wound site maintains better bioelectric coherence throughout healing, and the result is regenerative healing rather than fibrotic repair.

The clinical implications of this framework are significant and point toward a specific class of intervention. If regenerative competence is a bioelectric problem, if the limit of regeneration in non-regenerative organisms is primarily a limit of field coherence restoration, then the path to enhanced regeneration in humans does not necessarily require gene therapy to add regeneration genes that we supposedly lack, or stem cell transplantation to provide regeneration cell types that we supposedly cannot produce. It may require bioelectric intervention: ion channel-modulating drugs (electroceuticals) that push the wound site toward a regeneration-attractor bioelectric state; applied electric fields that impose voltage domain patterns consistent with the missing tissue; engineered gap junction topologies delivered through synthetic connexin-expressing biomaterials. These are all technically challenging, but they are mechanistically targeted in a way that previous regenerative medicine approaches have not been, because they are attacking the correct primary control variable.

There is also an evolutionary dimension worth acknowledging. The extraordinary regenerative capacity of planarians, hydra, and axolotls is not simply because they retained ancestral genetic programs that vertebrates discarded. It is because their bioelectric regulatory systems have remained tuned to rapidly and accurately re-establish field coherence after injury. The molecular machinery for regeneration is broadly conserved, the Wnt, BMP, Shh, and FGF pathways are present in virtually all metazoans, including mammals. What differs is the bioelectric regulatory layer that sits above those pathways and determines when and where they activate. Evolution of regenerative competence is substantially evolution of bioelectric regulation. And if that is true, then engineering regenerative competence, at least partially, means engineering bioelectric regulation, not necessarily adding new molecular programs from scratch.

The closing argument of this section, and the intellectual center of the paper, is this: biology spent decades searching for the regeneration gene. The search produced important discoveries, the conservation of developmental patterning genes, the identification of progenitor populations, the mapping of morphogen gradients in blastema tissue. But the

search operated on a fundamentally insufficient model of how positional information is encoded and read in regenerating tissue. The evidence increasingly demands that we acknowledge bioelectric field architecture as the primary control layer, the level at which spatial information is encoded, read, and acted on, with molecular biology as the implementation layer below it. Until we understand the bioelectric program, we will continue to be confused about why swapping out individual molecular components sometimes works and often doesn't. The field is the program. The molecules are the hardware. You need both, but you need to understand the program to debug the computer.

SECTION 10

CONCLUSION, WHAT THIS MEANS AND WHERE IT GOES

The argument of this paper can be stated as a sequence, because the biology unfolds as a sequence, and each step follows from the one before it in a way that is mechanistically traceable rather than hand-waved. Injury causes field collapse: the transepithelial potential drops instantly at the wound edge, a lateral wound electric field forms pointing toward the wound center, calcium waves propagate outward, and the depolarization zone at the wound site is established as the first spatial signal available to the surrounding tissue. Wound field formation provides the coordinates: the field tells cells where the damage is, polarizes them toward the wound through receptor redistribution and calcium asymmetry, and drives electrotaxis-mediated migration before any diffusible signal has organized enough to be read. Polarity encoding resolves the symmetry-breaking question: differential voltage states at the two wound faces in a bisected animal determine which bioelectric attractor each blastema settles into, and the downstream gene expression asymmetry that specifies head versus tail, proximal versus distal, dorsal versus ventral, follows from that voltage state. Voltage domain re-establishment provides fine-grained positional identity: coherent electrical territories form in the blastema, their boundaries maintained by differential ion pump and channel expression and by gap junction coupling, and these domains precede and predict the organ and tissue territories that will form within them.

Gap junction network reconstruction re-establishes the electrical integration required to maintain domain coherence and to propagate positional information through the regenerating tissue. Electrotaxis-driven migration continues to populate the repair site as the wound field evolves and the blastema expands. And finally, developmental template replay, driven by the restored bioelectric field architecture, executed by re-expressed developmental patterning gene networks, and terminated when field coherence is achieved, rebuilds the missing structure. Regeneration complete.

This framework is not a rejection of molecular biology. It is not an argument that genes do not matter, that morphogens are unimportant, or that stem cells are irrelevant. It is something more precise and more useful: it is a higher-order explanatory framework that specifies when and where the molecular machinery operates, and that explains why the same molecular players produce different outcomes in different contexts. The reason that introducing Wnt ligands to a wound site in a mammal does not reliably produce limb regeneration, even though Wnt is essential for limb regeneration in axolotls, is not that the mammalian cells cannot respond to Wnt, they can. It is that the spatial context into which Wnt is being delivered is a scar-attractor bioelectric environment, not a regeneration-attractor environment, and so the cells that receive Wnt signaling cannot interpret it as a regenerative positional instruction. Provide the correct bioelectric context first, and then deliver the molecular signal, and the outcome may be radically different. This is the testable hypothesis that the bioelectric framework generates, and there is already meaningful experimental evidence supporting it.

The testable predictions of this framework are specific enough to be scientifically productive rather than vacuously broad. First: manipulating bioelectric state before or in conjunction with other regenerative interventions should modulate regenerative outcome in dose-dependent, mechanistically predictable ways. Depolarizing a wound site should push it toward regeneration-incompetent attractor states; hyperpolarizing it with the correct ion channel complement should push it toward regeneration-competent states. These predictions can be tested with existing pharmacological and optogenetic tools.

Second: organisms or tissues engineered to maintain bioelectric field coherence after injury, through overexpression of the relevant connexin isoforms, ion pumps, and channels, should show enhanced regenerative outcomes even without additional molecular interventions. Third: the limit of regeneration in non-regenerative organisms should correlate directly with the limit of voltage domain restoration at the wound site, and this correlation should be measurable using existing voltage-imaging methodology applied to healing wounds over time.

The open questions that remain are genuine and significant, and they deserve honest acknowledgment. How exactly does V_{mem} modulate transcription factor activity with sufficient spatial and temporal resolution to drive precise patterning? The broad pathways are known, calcium-NFAT, cAMP-CREB, pH-mediated enzyme activity, but the quantitative relationship between specific V_{mem} values and specific transcriptional outcomes in specific cell types is not well mapped. What is the full vocabulary of voltage domain states? We know that distinct domains exist and that they correspond to distinct tissue identities, but the complete mapping of V_{mem} domain states to morphogenetic outcomes, the bioelectric code, has not been systematically established. Can gap junction network topology be engineered in situ in a living organism, with sufficient precision to guide regenerative outcomes? Early results with connexin-expressing biomaterials are encouraging but not yet at the precision required for complex structure regeneration. And how do we deliver bioelectric interventions, applied fields, ion channel modulators, connexin-engineering, without producing systemic effects that would limit therapeutic use? These are solvable engineering problems, but they are not trivial, and the field's track record on translation from animal models to human therapy demands honest caution.

None of these open questions undermine the central framework. They are the research agenda that the framework generates, specific, mechanistically grounded questions that can be addressed with current and foreseeable technology. That is what a good scientific framework does: it does not close questions, it opens better ones. The hypothesis that regeneration is primarily a field coherence restoration problem is not established with the

finality that the laws of thermodynamics are established. But the weight of evidence across multiple experimental systems, across diverse model organisms, and across multiple levels of biological organization, from single-cell electrophysiology to tissue-level bioelectric imaging to whole-organism regenerative outcome, is increasingly consistent with this framework and increasingly inconsistent with the view that regeneration is primarily a genetic or stem-cell problem at its core.

Regeneration research is at an inflection point. The tools to image, measure, and manipulate bioelectric fields in living tissue have improved dramatically over the past decade: genetically encoded voltage indicators, optogenetic ion channel control, voltage-sensitive dyes with sufficient sensitivity to image individual cell-level V_{mem} in intact tissues, and vibrating probe electrodes capable of mapping ion current distributions with high spatial resolution. These tools are now being applied to regeneration models with increasing sophistication, and the picture that emerges from that application is consistently one in which bioelectric architecture is an early, organizing, causally upstream feature of the regenerative process rather than a correlate or a downstream readout. The field is moving from demonstrating that bioelectric signals correlate with regenerative outcomes to demonstrating that they cause them, and that distinction is the line between scientific curiosity and scientific explanation.

The decade ahead will be shaped by how well regeneration scientists learn to read, write, and restore the body's electrical map. Reading it means developing the imaging and measurement tools to characterize bioelectric field architecture in regenerating tissues with sufficient resolution to distinguish attractor states. Writing it means developing the pharmacological, optogenetic, and bioelectrical intervention tools to impose specific voltage domain patterns on wound sites in a targeted, controllable way. Restoring it means translating those tools into clinical settings, wound care, peripheral nerve repair, spinal cord injury, limb reconstruction, where the bioelectric framework predicts benefits that conventional molecular approaches have failed to deliver. The biology is ready. The framework is coherent. The question now is whether the field of regenerative medicine is

ready to absorb a new primary explanatory variable and build a research and clinical agenda around it. The evidence says it should be. The body has been running this electrical program since long before we had the technology to see it. It is time we learned to read it seriously.